

Validation scorecard: twelve executed experiments

E1	Leave-one-city-out CV <i>[Calibration]</i>	ρ_{LOO} value = 0.52 threshold = 0.30	PASS
E2	Weight-space LOO bootstrap <i>[Calibration]</i>	Top-10 retention value = 0.90 threshold = 0.50	PASS
E3	Behavioural vs eco-counters <i>[Construct]</i>	Partial R^2 net size value = 0.24 threshold = 0.10	PASS
E4	Exposure-adjusted safety <i>[Construct]</i>	ρ_5 shift (raw $\rightarrow$ adj.) value = 0.92 threshold = 0.50	QUALIFIED
E5	Within-city station bootstrap <i>[Geometry]</i>	Kendall τ (median) value = 0.85 threshold = 0.70	PASS
E6	IES specification sweep <i>[Equity]</i>	Robust deserts $P \geq 0.75$ value = 0.83 threshold = 0.50	PASS
E7	Sobol/Jansen variance decomposition <i>[Geometry]</i>	$\sum_k S_k$ (score) value = 0.98 threshold = 0.80	PASS
E8	Spatial autocorrelation <i>[Panel]</i>	Moran I value = 0.03 threshold = 0.10	PASS
E9	Bayesian IES posterior <i>[Equity]</i>	$P(\beta_{\text{velo}} > 0)$ value = 1.00 threshold = 0.95	PASS
E10	Per-city IMD bootstrap CI <i>[Panel]</i>	Top-3 CI-distinct value = 1.00 threshold = 0.80	PASS
E11	Parametric buffer-radius sweep <i>[Geometry]</i>	min Kendall τ (150-500m) value = 0.92 threshold = 0.70	PASS
E12	Cook-style leverage <i>[Panel]</i>	$\max \Delta \bar{\rho} $ value = 0.07 threshold = 0.10	QUALIFIED
E13	k-means typology <i>[Panel]</i>	Silhouette ($k = 4$) value = 0.30 threshold = 0.20	PASS